

WADI AL DAWASIR, Saudi Arabia

WMO: 410610

Lat: 20.504N

Lon: 45.200E

Elev: 629

StdP: 94.00

Time Zone: 3.00 (E03)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	9.8	N/A	N/A	N/A	N/A	N/A	N/A	10.2	18.5	9.3	18.6	2.9	110	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	44.9	N/A	44.1	N/A	43.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.6	0

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	(o)	(p)
9.6	8.6	7.6	5.2	46.4	1.6	0.9	4.0	47.1	3.0	47.6	2.1	48.1	0.9	48.8	(4)	(5)
			DB													
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	17.7	20.9	24.8	29.2	33.4	35.3	36.5	36.3	33.0	28.0	23.0	18.8
	DBStd	7.05	3.03	3.31	3.11	2.70	1.74	1.33	1.49	1.27	1.84	1.91	2.30	2.93
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	95	49	13	1	0	0	0	0	0	0	0	2	31
	CDD10.0	6611	238	305	459	576	726	759	822	815	691	559	389	271
	CDD18.3	3664	29	84	202	326	468	509	563	557	441	301	141	43
	CDH23.3	57828	342	917	2334	4542	7763	8936	10095	9892	7215	3969	1381	444
CDH26.7	39254	82	366	1165	2735	5368	6556	7618	7416	4968	2324	551	107	
Wind	WSAvg	3.7	4.2	4.4	4.6	4.2	3.5	3.1	3.4	3.2	2.9	3.0	3.9	4.0
Precipitation	PrecAvg	35	2	2	12	13	3	0	1	0	0	0	1	1
	PrecMax	88	20	18	47	63	47	1	7	5	1	8	14	14
	PrecMin	8	0	0	6	0	0	0	0	0	0	0	0	0
	PrecStd	23	5	5	10	18	10	0	1	1	0	2	4	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	35.2	38.8	40.8	43.2	45.1	46.1	45.2	43.9	39.1	34.1	31.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	28.9	32.9	36.1	39.2	42.2	44.1	45.1	44.2	42.8	37.9	32.6	29.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	27.0	31.0	34.2	38.1	41.2	43.2	44.1	43.9	41.8	36.8	31.2	27.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	25.0	29.0	32.8	36.9	40.8	42.8	43.2	43.1	40.8	35.8	30.1	26.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.0	13.7	13.7	13.9	14.9	14.2	14.3	15.2	15.0	13.5	13.4
		MCDBR	15.8	16.7	16.0	15.2	15.1	15.6	14.8	15.0	15.9	15.6	14.8	14.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.484	0.528	0.597	0.625	0.640	0.664	0.682	0.670	0.574	0.467	0.467	0.451	
	taud	1.842	1.730	1.595	1.578	1.533	1.481	1.492	1.510	1.655	1.872	1.882	1.909	
	Ebn at Noon	774	767	733	720	704	681	669	680	741	811	783	788	
	Edh at Noon	192	227	270	278	288	301	298	294	251	194	182	174	
All-Sky Solar Radiation	RadAvg	4.90	5.74	6.45	6.95	7.43	7.57	7.22	7.09	6.81	6.29	5.25	4.75	
	RadStd	0.29	0.21	0.26	0.28	0.28	0.21	0.23	0.15	0.11	0.14	0.15	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	+0.40	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page